A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Renewable energy technologies - Set 2

Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 196-217

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 4.2 / 0.7 = 6 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low **1** In winter days are very short / the Sun does not rise, so light levels are extremely low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight onto the absorber **1** Clouds, smoke and dust scatter the light so it cannot be focused and intensified **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	1 mark for the point below. **1** Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

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Question	Marking guidance	Marks
01.5	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Problems:** intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • **Photothermal:** evacuated tubes reduce conduction / convection losses – thermal stores (water, sand, concrete, molten salt) allow later use – passive solar architecture, brise soleil, light tubes – heat pumps deliver up to four times the energy used • **Photovoltaic developments:** multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • **Integration:** grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • **Evaluation:** strong in sunny deserts / low latitudes, weaker in high latitudes in winter – cost falling – manufacturing impacts – no single technology solves all problems – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (9 marks)	
	Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.	
	New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.	
	Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.	
	Weaker answer - Level 1 (2 marks)	
	Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.	
	What makes the difference	
	• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.	
	• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.	
	• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	

Do not write outside the box